

Spectral Entropy as a Multiscale Diagnostic of Correlation Structure in Decimal Digit Sequences

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Abstract

We compare Shannon entropy and spectral (hydrodynamic) entropy as diagnostics of structure in decimal digit sequences of mathematical constants and algorithmic generators. Shannon entropy, which measures local digit-frequency uniformity, rapidly saturates to its maximum value $\log_2 10 \approx 3.322$ bits for all sequences examined, including π , e , $\sqrt{2}$, the Champernowne constant, pseudorandom generators, and periodic controls.

Spectral entropy, computed from the Fourier power spectrum, probes correlations across multiple length scales. For digit lengths up to 1.5×10^6 , π , e , $\sqrt{2}$, uniform random sequences, and the logistic map at $r = 4$ exhibit scaling $S_H \sim \alpha \log_{10} N$ with $\alpha \approx 3.32$, consistent with the theoretical prediction for uniform Fourier mode population. In contrast, structurally constrained sequences such as Champernowne ($\alpha \approx 1.24$), Thue–Morse ($\alpha \approx 2.43$), and periodic sequences ($\alpha \approx 0$) display suppressed scaling. The De Bruijn sequence uniquely exhibits scale-dependent growth captured by a quadratic model.

These results indicate that spectral entropy distinguishes deterministic, recursive, pseudorandom, and chaotic constructions even when digit frequencies are asymptotically uniform. Within the tested range, spectral-entropy scaling appears empirically independent of classical digit normality. Potential applications include complementary diagnostics for pseudorandom-number generators and complexity analysis of symbolic sequences.

1 Introduction

The decimal expansions of mathematical constants provide a classical testing ground for notions of randomness and normality. A real number is normal in base 10 if every digit and every finite block of digits occurs with equal limiting frequency [1]. Although π and e are widely conjectured to be normal [2], no proof currently exists, and empirical investigation remains informative.

Most computational studies focus on local statistics such as digit frequencies, block distributions, or classical randomness test suites [3, 4, 5]. Such diagnostics probe distributional uniformity but are insensitive to ordering beyond short blocks. Consequently, sequences that are provably normal yet algorithmically simple, such as the Champernowne constant [6], may be indistinguishable from pseudorandom sequences under purely local measures.

1.1 Motivation: Separating Normality from Algorithmic Complexity

A critical but underappreciated distinction exists between:

1. **Digit normality:** uniform distribution of symbols (a local property), and
2. **Algorithmic complexity:** computational irreducibility (a global property).

The Champernowne constant $C_{10} = 0.123456789101112\dots$, formed by concatenating integers, is *provably normal* [6] yet exhibits obvious deterministic structure—it can be generated by a trivial $O(\log n)$

program. Conversely, algebraic irrationals such as $\sqrt{2}$ have *unknown* normality status in base 10 [7, 8]—neither normality nor non-normality has been established—yet they have no simple closed-form expression for their n -th digit, requiring full-precision root extraction.

This motivates our central question: *Can spectral analysis distinguish algorithmically simple sequences (like Champernowne) from sequences lacking exploitable spectral structure (like π , e , or $\sqrt{2}$), independent of digit normality?*

1.2 Spectral Entropy: A Multiscale Diagnostic

Shannon entropy $H = -\sum p_d \log_2 p_d$ quantifies uncertainty in digit frequencies but is insensitive to ordering—it is permutation-invariant [9, 10]. Two sequences with identical digit distributions yield identical Shannon entropy regardless of underlying structure, exposing the fundamental limitation of local measures.

Spectral entropy, computed from the Fourier power spectrum, probes correlations across all length scales simultaneously [11]. Originally developed for signal processing and adapted to turbulent flows as “hydrodynamic entropy” [12, 13], it measures how uniformly energy (variance) distributes across independent Fourier modes. Throughout this paper, we use the terms “spectral entropy” and “hydrodynamic entropy” interchangeably, following the convention of [12].

For a sequence of length N , approximately $N/2$ independent Fourier modes exist due to Hermitian symmetry [11]. If energy distributes uniformly across these modes (white noise), spectral entropy grows as:

$$S_H = \log_2(N/2) = \log_2 N - 1 = \frac{\ln 10}{\ln 2} \log_{10} N - 1, \quad (1)$$

where the last step uses the change-of-base identity $\log_2 N = (\ln 10 / \ln 2) \log_{10} N$. Fitting $S_H = \alpha \log_{10} N + \beta$, we therefore expect:

$$\alpha_{\text{theory}} = \frac{\ln 10}{\ln 2} \approx 3.322 \quad (2)$$

for truly random sequences with uniform Fourier mode population. We emphasise that S_H is not normalized by $\log_2(N/2)$; all theoretical comparisons are therefore made with the unnormalised expression in Eq. (4).

1.3 Key Contributions

This study makes four primary contributions:

1. **Theoretical agreement.** π , e , $\sqrt{2}$, and uniform random sequences all exhibit $\alpha \approx 3.32$ (Table 1), consistent with Eq. (2), indicating that spectral entropy can serve as a diagnostic for the absence of exploitable spectral correlations.
2. **Independence of normality and spectral complexity at tested scales.** The provably normal Champernowne constant [6] shows $\alpha \approx 1.24$ (Table 1), substantially lower than $\sqrt{2}$ (normality open [7]), π , or e —indicating that normality and spectral complexity need not coincide at these scales (Table 2).
3. **Theoretical framework.** Derivation of $\alpha_{\text{theory}} = \ln(10)/\ln(2)$ from uniform Fourier mode population (Section 2.5), with interpretation via the Wiener–Khinchin theorem.
4. **Graded complexity hierarchy.** Spectral entropy produces a continuous ranking across random, chaotic, pseudorandom, recursive, and deterministic sequences (Section 3.5), including a characterisation of scale-dependent spectral entropy growth in De Bruijn sequences [16] (Section 3.6.1).

2 Methodology

2.1 Digit Extraction and Data Sources

We analyse decimal digit sequences from six primary sources:

- π : first 1.5×10^6 digits via the Chudnovsky algorithm;
- e : first 1.5×10^6 digits via Taylor series expansion;
- $\sqrt{2}$: first 1.5×10^6 digits via high-precision arithmetic;
- Champernowne C_{10} : constructed by concatenating $1, 2, 3, \dots$ [6];
- Uniform random: 10 independent realisations from the discrete uniform distribution on $\{0, \dots, 9\}$;
- Periodic: repeating sequence $(0, 1, 2, \dots, 9)$.

An extended benchmark (Section 3.5) adds four further constructions:

- De Bruijn sequence: base-10, order 7, cycle length $L = 10^7$ [16];
- Linear Congruential Generator (LCG): $x_{n+1} = ax_n + c \pmod{m}$ [15];
- Logistic map at $r = 4$ (fully chaotic regime [17]), digits extracted from iterates;
- Thue–Morse substitution sequence [18], mapped to base-10 digits.

Logarithmic sampling is employed throughout: $N = 10^x$, $x \in [1, 6.18]$, using 1 000 geometrically spaced points to probe five orders of magnitude efficiently.

2.2 Shannon Entropy

For N digits with empirical probabilities $p(d) = n_d/N$, Shannon entropy is [9]:

$$H = - \sum_{d=0}^9 p(d) \log_2 p(d). \quad (3)$$

The maximum $H_{\max} = \log_2 10 \approx 3.322$ bits is attained when all digits are equally frequent [10].

2.3 Spectral Entropy

Treating the digit sequence $\{u_n\}_{n=1}^N$ as a discrete signal [11]:

1. Centre: $\tilde{u}_n = u_n - \bar{u}$.
2. Discrete Fourier transform: $\psi_k = \sum_{n=1}^N \tilde{u}_n e^{-2\pi i(k-1)(n-1)/N}$.
3. Modal energy: $E_k = \frac{1}{2} |\psi_k|^2$ for $k = 1, \dots, \lfloor N/2 \rfloor$.
4. Normalise: $p_k = E_k / \sum_j E_j$.
5. Spectral entropy: $S_H = - \sum_k p_k \log_2 p_k$ [12].

Physical interpretation. For uncorrelated sequences, each Fourier mode carries independent information; as N increases, more modes become available and S_H grows. Deterministic structure concentrates energy in specific modes, suppressing S_H growth [13].

2.4 The Spectral Advantage: Permutation-Invariance vs. Order-Sensitivity

Shannon entropy is a function of the probability mass function alone and is therefore *permutation-invariant* [10]: for any sequence S , $H(S) = H(\text{sort}(S))$. This is why it fails to distinguish the Champernowne constant from a random sequence—it discards all ordering information.

Spectral entropy, by contrast, is *order-sensitive* via the Wiener–Khinchin theorem [11]: the power spectrum equals the Fourier transform of the autocorrelation $R_{uu}(\tau)$, so S_H captures the decay of correlations across all lags. For an algorithmically simple sequence like Champernowne, the predictable transitions manifest as long-range correlations that concentrate power in specific frequencies, suppressing entropy growth.

We note that block entropy H_n (over n -gram frequencies) and conditional entropy provide intermediate diagnostics sensitive to some correlations [10]. However, block entropy becomes impractical beyond $n \sim 5$ for base-10 sequences (requiring estimation of $10^5 = 100,000$ n -gram probabilities), whereas spectral entropy probes all correlation scales up to $N/2$ via a single $O(N \log N)$ FFT.

2.5 Theoretical Scaling Derivation

For a truly random sequence with N digits and uniform mode population:

- Number of independent Fourier modes: $M = N/2$ (Hermitian symmetry [11]).
- Uniform energy: $p_k = 1/M = 2/N$ for all k .
- Spectral entropy:

$$\begin{aligned} S_H &= - \sum_{k=1}^{N/2} \frac{2}{N} \log_2 \left(\frac{2}{N} \right) = - \frac{N}{2} \cdot \frac{2}{N} \log_2 \left(\frac{2}{N} \right) \\ &= - \log_2(2/N) = \log_2(N/2) = \log_2 N - 1. \end{aligned} \quad (4)$$

- Converting via $\log_2 N = (\ln 10 / \ln 2) \log_{10} N$:

$$S_H = \frac{\ln 10}{\ln 2} \log_{10} N - 1 \approx 3.322 \log_{10} N - 1. \quad (5)$$

- Therefore $\alpha_{\text{theory}} = \ln 10 / \ln 2 \approx 3.322$ (Eq. 2).

Sequences with $\alpha \approx 3.32$ thus exhibit uniform mode population, consistent with the absence of exploitable spectral correlations (confirmed empirically in Table 1).

2.6 Statistical Analysis

We fit $S_H(N) = \alpha \log_{10} N + \beta$ via weighted least-squares over $10^3 \leq N \leq 10^6$, following standard linear regression methodology [20]. For random sequences, we report mean $\pm$ standard deviation across 10 realisations. Goodness of fit is assessed via R^2 , and 95% confidence intervals on α are obtained from the regression covariance matrix [20]. Sensitivity to the fitting range is tested by varying the lower bound from 10^2 to 10^4 .

For sequences exhibiting nonlinear spectral-entropy growth (Section 3.6.1), we additionally fit a quadratic model $S_H(x) = ax^2 + bx + c$ with $x = \log_{10} N$ and report the scale-dependent effective exponent $\alpha_{\text{eff}}(N) = dS_H/d\log_{10} N = 2ax + b$.

3 Results

3.1 Digit Frequency Convergence

3.1.1 Mathematical Constants: π , e , and $\sqrt{2}$

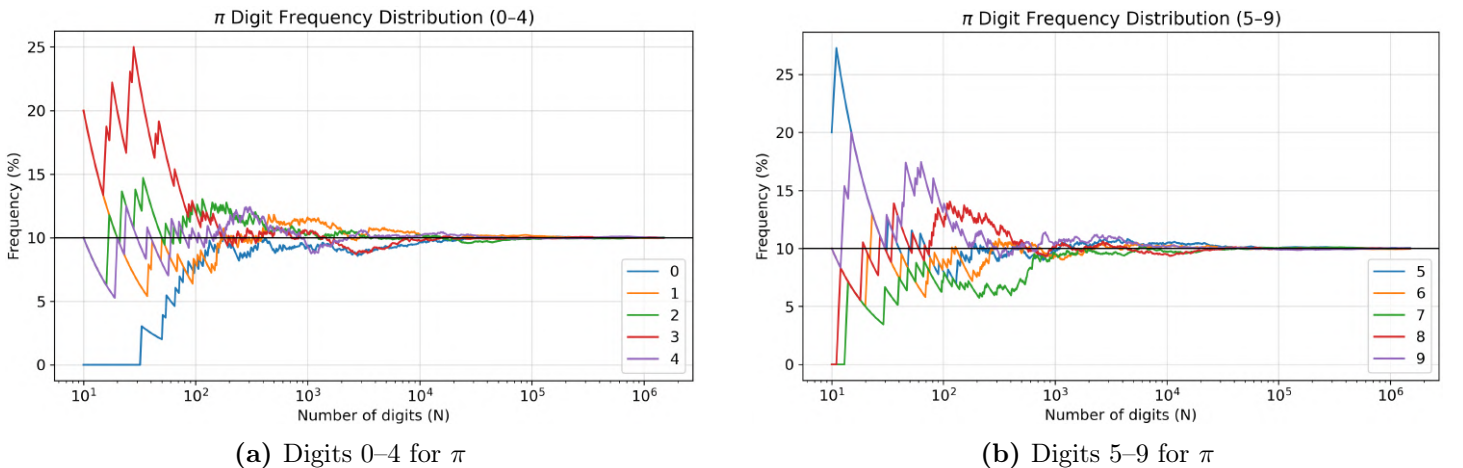
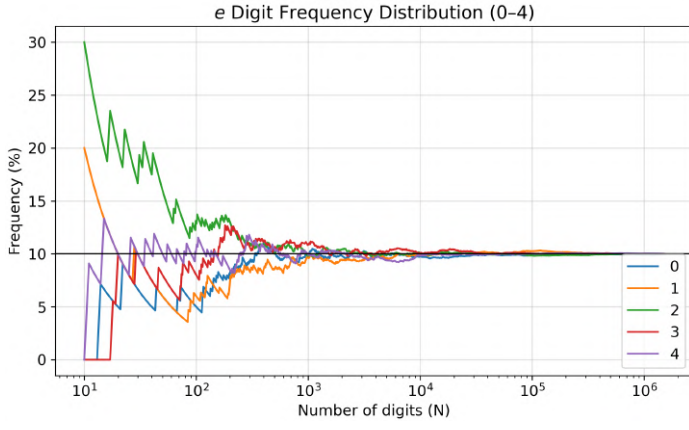
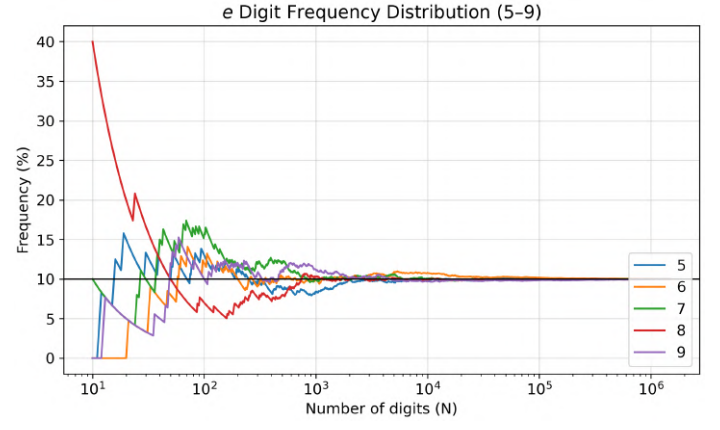


Figure 1: Digit frequency convergence for π . All digits converge toward 10% with decreasing fluctuations as N increases, consistent with conjectured normality [2].

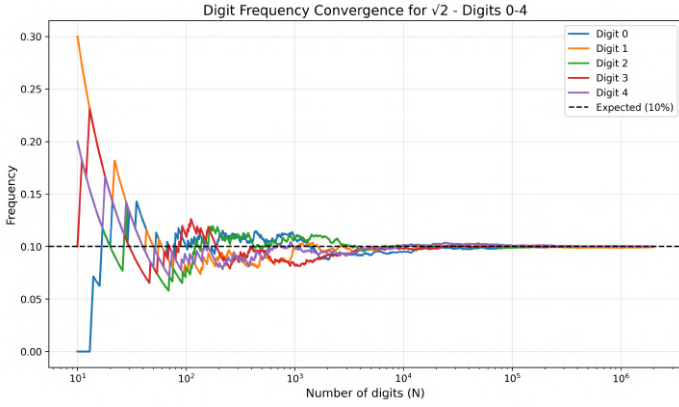


(a) Digits 0–4 for e

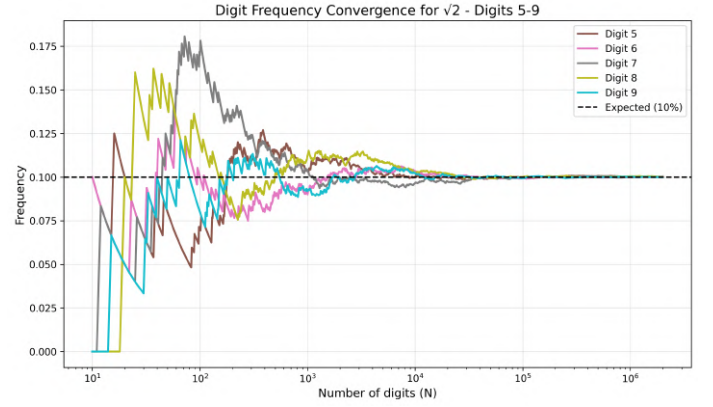


(b) Digits 5–9 for e

Figure 2: Digit frequency convergence for e . Similar to π , all digits approach uniform distribution [3].



(a) Digits 0–4 for $\sqrt{2}$

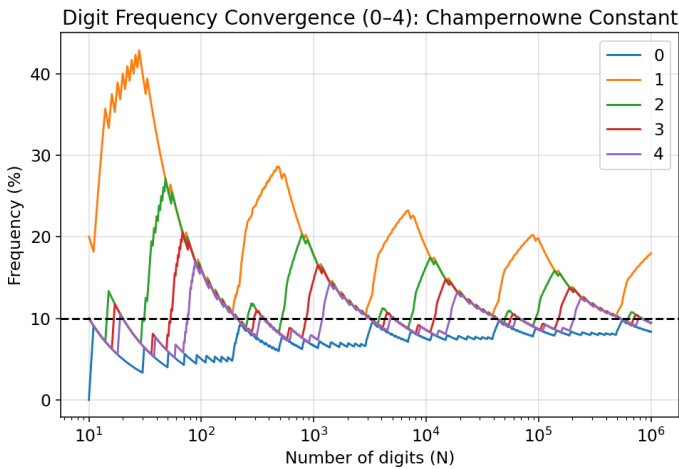


(b) Digits 5–9 for $\sqrt{2}$

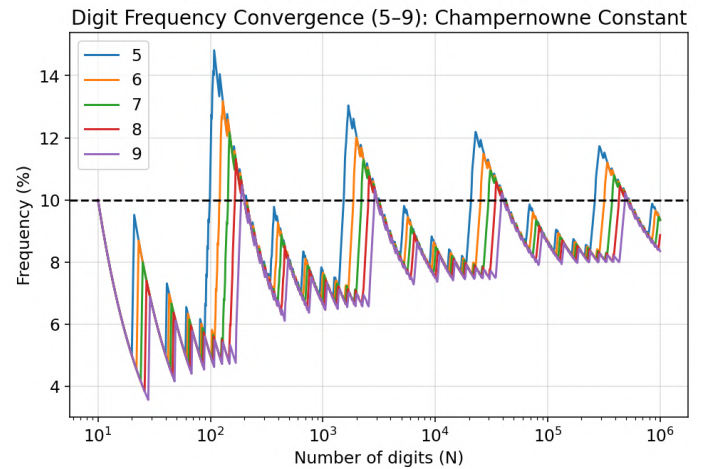
Figure 3: Digit frequency convergence for $\sqrt{2}$. Frequencies converge toward 10% at tested scales, consistent with (but not proving) normality. The normality of $\sqrt{2}$ in base 10 remains open [7].

Figures 1–3 show that all three constants converge toward uniform digit frequencies within $N \leq 1.5 \times 10^6$. Fluctuations decrease as $N^{-1/2}$, consistent with binomial statistics.

3.1.2 Reference Sequences: Champernowne, Random, and Periodic

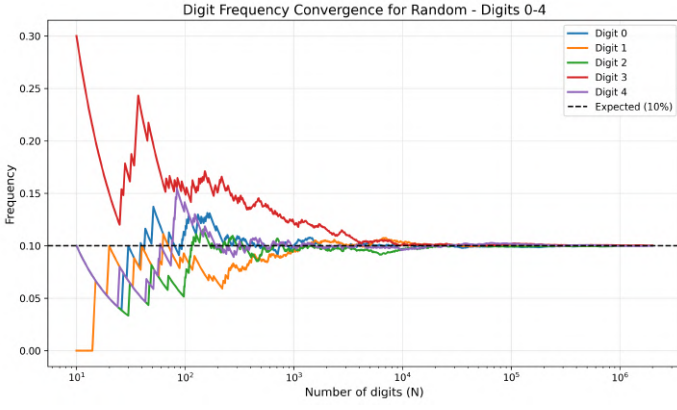


(a) Digits 0–4 for Champernowne

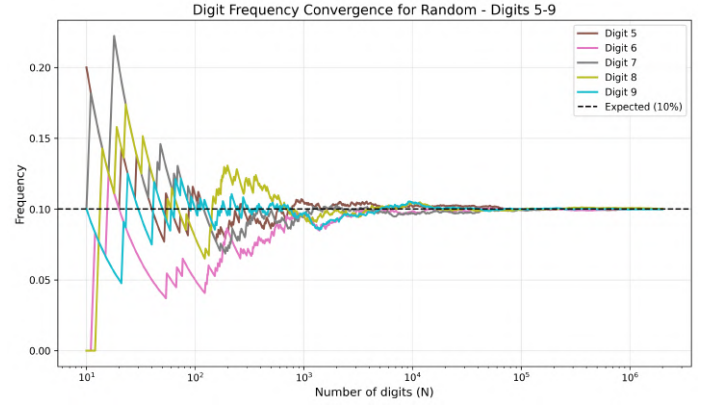


(b) Digits 5–9 for Champernowne

Figure 4: Digit frequency convergence for the Champernowne constant. Despite proven normality [6], oscillations reveal the block structure from integer concatenation. Digit 1 shows peaks reaching $\sim 40\%$ at scales corresponding to ranges $[10^k, 2 \times 10^k)$.

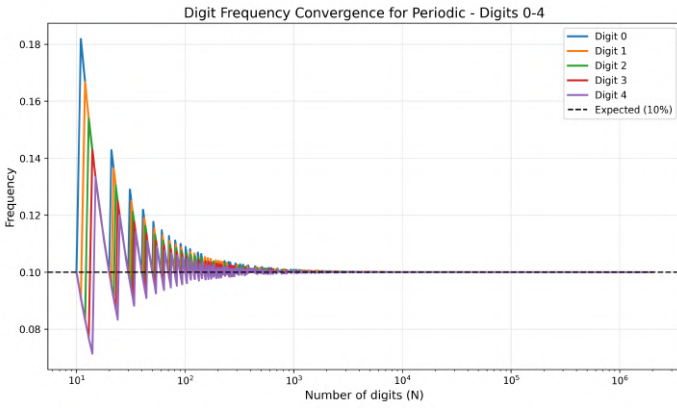


(a) Digits 0–4 for Random

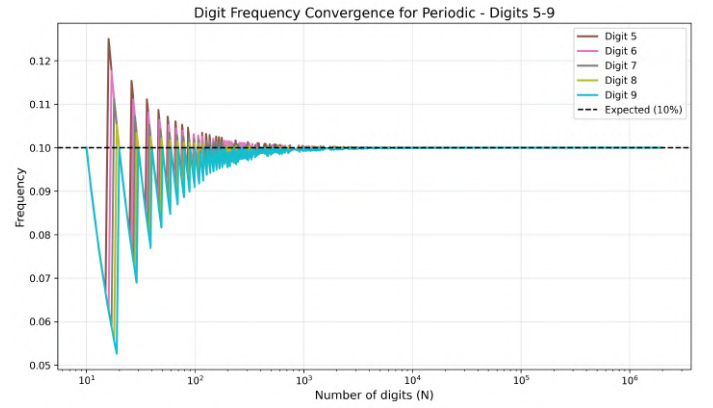


(b) Digits 5–9 for Random

Figure 5: Digit frequency convergence for a uniform random sequence. All digits converge toward 10% with statistical fluctuations consistent with a binomial distribution.



(a) Digits 0–4 for Periodic



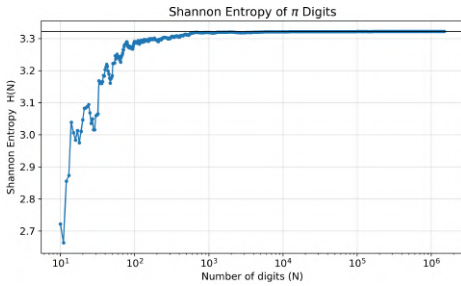
(b) Digits 5–9 for Periodic

Figure 6: Digit frequency for the periodic sequence $(0, 1, 2, \dots, 9)$ repeating. All digits maintain exactly 10% by construction.

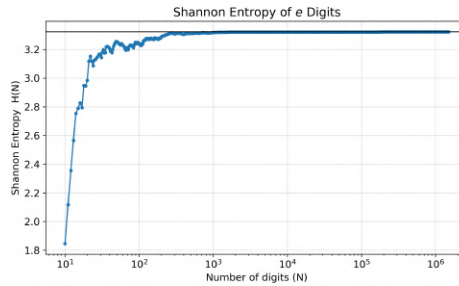
The reference sequences (Figures 4–6) confirm that digit-level uniformity alone cannot distinguish algorithmic complexity. Champernowne, despite proven normality [6], exhibits oscillatory deviations at finite N , while the trivial periodic sequence maintains perfect uniformity by design.

3.2 Shannon Entropy: Local Measure

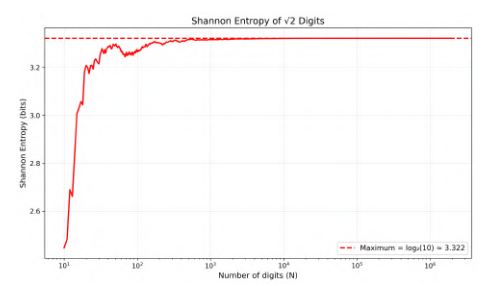
3.2.1 Mathematical Constants



(a) π



(b) e



(c) $\sqrt{2}$

Figure 7: Shannon entropy saturates to $H_{\max} = \log_2 10 \approx 3.322$ bits by $N \approx 10^4$ for all three constants. $\sqrt{2}$ (normality unknown [7]) exhibits identical saturation to π and e (conjectured normal [2]), illustrating that Shannon entropy detects only local digit uniformity.

3.2.2 Reference Sequences

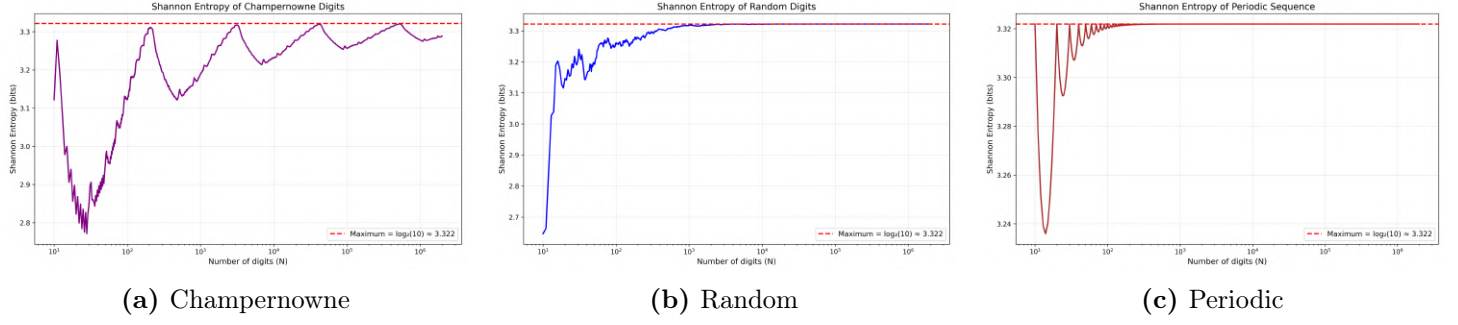


Figure 8: Shannon entropy for reference sequences. Champernowne (provably normal [6]), Random, and Periodic all converge to $\log_2 10 \approx 3.322$ bits, illustrating Shannon entropy’s insensitivity to long-range correlations—the limitation motivating spectral analysis.

3.3 Spectral Entropy: Multiscale Measure

3.3.1 Mathematical Constants

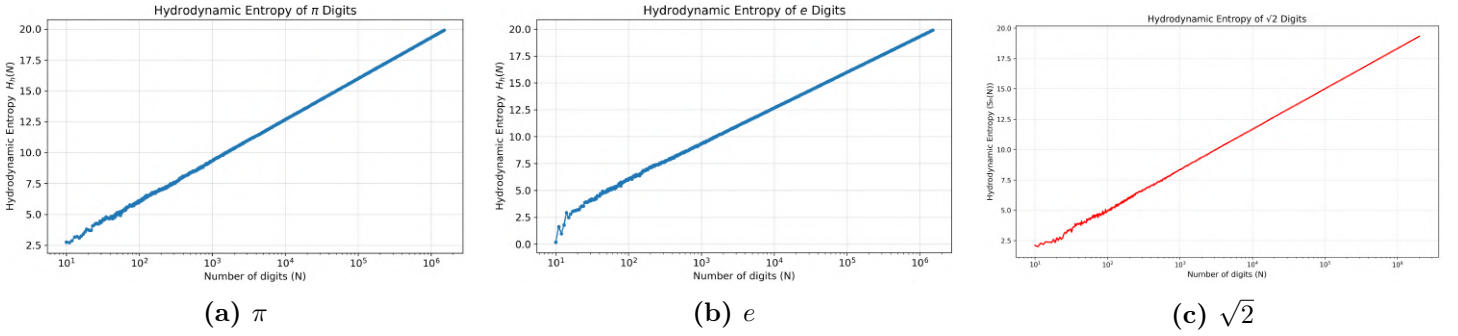


Figure 9: Spectral entropy for π , e , and $\sqrt{2}$ grows logarithmically with $\alpha \approx 3.32$ (Table 1), matching the theoretical prediction of Eq. (2). Despite different mathematical origins (transcendental vs. algebraic), all three exhibit comparable spectral scaling.

In contrast to Shannon entropy, spectral entropy clearly differentiates among sequences. Figure 9 shows that π , e , and $\sqrt{2}$ all yield $\alpha \approx 3.32$, matching the white-noise prediction of Eq. (2).

3.3.2 Reference Sequences

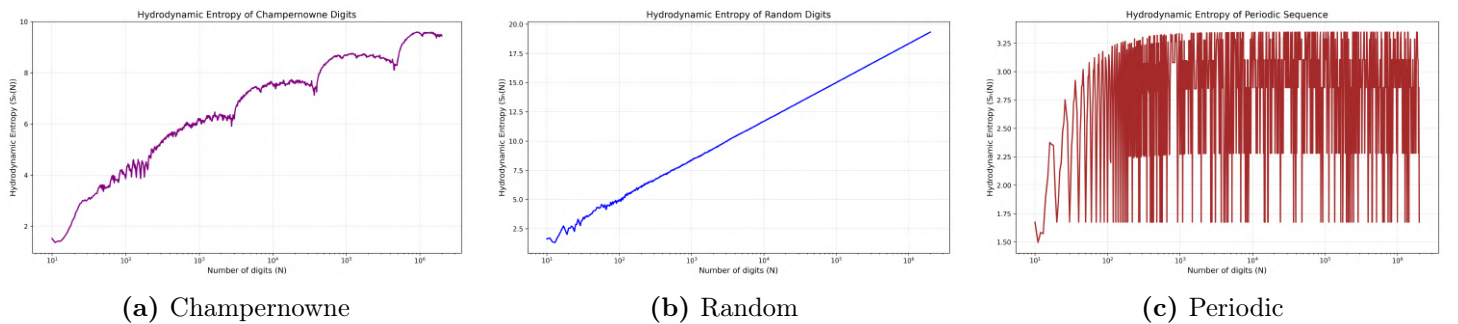
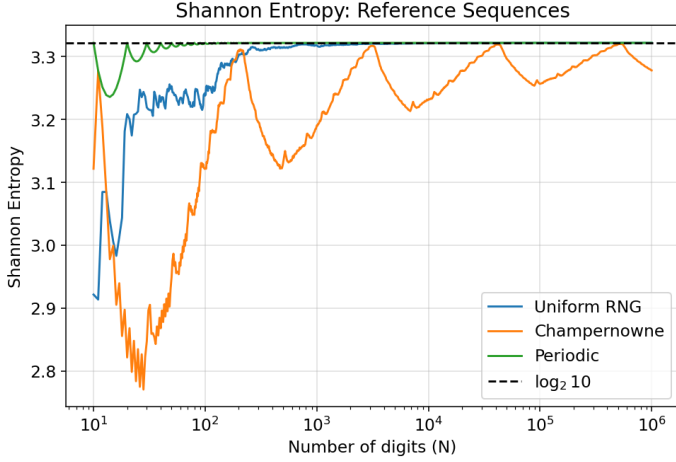


Figure 10: Spectral entropy for reference sequences reveals a clear hierarchy: Random ($\alpha \approx 3.32$), Champernowne ($\alpha \approx 1.24$), and Periodic ($\alpha \approx 0$). Quantitative values in Table 1.

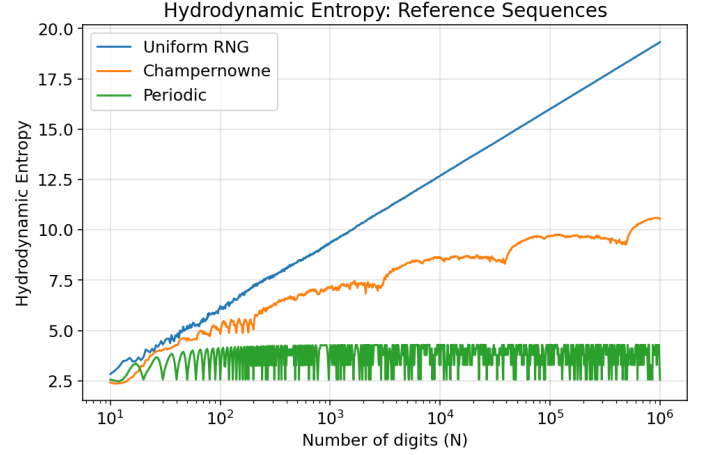
Figure 10 confirms that spectral entropy separates sequences by their correlation structure, independent of digit-level normality. Champernowne, despite proven normality [6], exhibits substantially lower spectral-entropy growth than Random.

3.4 Comprehensive Comparisons

3.4.1 Shannon vs. Spectral Entropy



(a) Shannon entropy

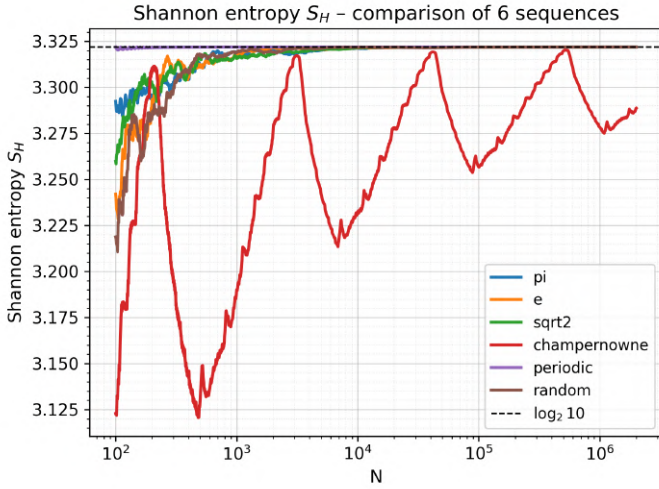


(b) Spectral entropy

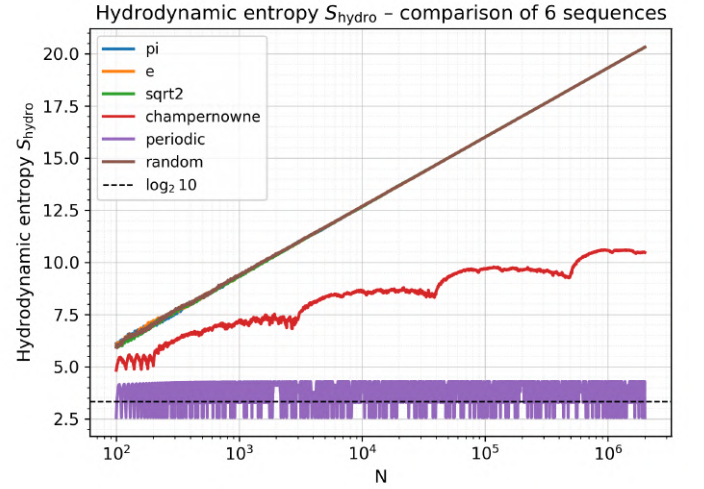
Figure 11: Direct comparison of entropy measures for reference sequences. (a) Shannon entropy: all sequences with uniform digits converge to $\log_2 10 \approx 3.322$ bits. (b) Spectral entropy: hierarchical separation emerges.

Figure 11 illustrates the core distinction. Shannon entropy discards ordering information and conflates all sequences with uniform digits; spectral entropy resolves their multiscale correlation structure.

3.4.2 Complete Hierarchy



(a) Shannon entropy



(b) Spectral entropy

Figure 12: Six-sequence hierarchy. $\sqrt{2}$ (algebraic, normality unknown [7]) exhibits $\alpha \approx 3.32$, matching π , e , and Random. Champernowne (provably normal [6]) shows $\alpha \approx 1.24$. Periodic shows trivial structure ($\alpha \approx 0$). All values from Table 1.

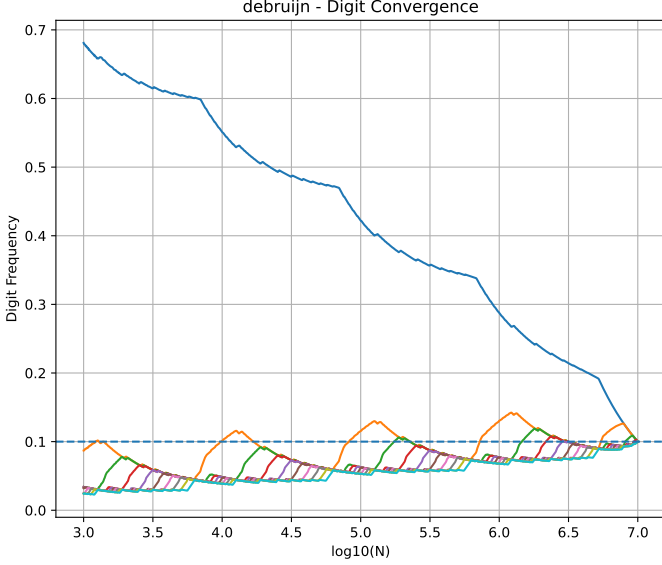
Figure 12 presents the complete hierarchy (quantified in Table 1):

1. **Random**, π , e , $\sqrt{2}$: $\alpha \approx 3.32$ (consistent with Eq. 2).
2. **Champernowne**: deterministic concatenation, $\alpha \approx 1.24$.
3. **Periodic**: trivial pattern, $\alpha \approx 0$.

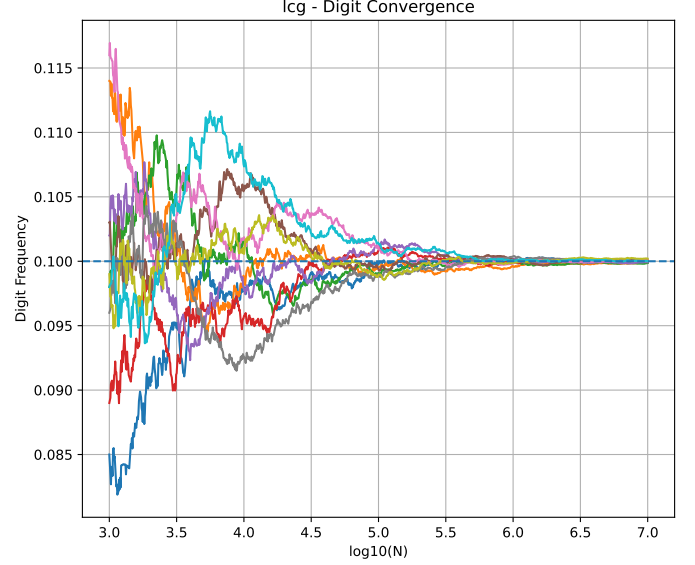
3.5 Extended Generator Benchmark

To test generality, we analyse four additional sequences: the De Bruijn sequence [16], an LCG [15], the logistic map at $r = 4$ [17], and the Thue–Morse sequence [18]. These span deterministic finite-cycle systems, classical pseudorandom generators, chaotic dynamics, and recursively structured aperiodic sequences.

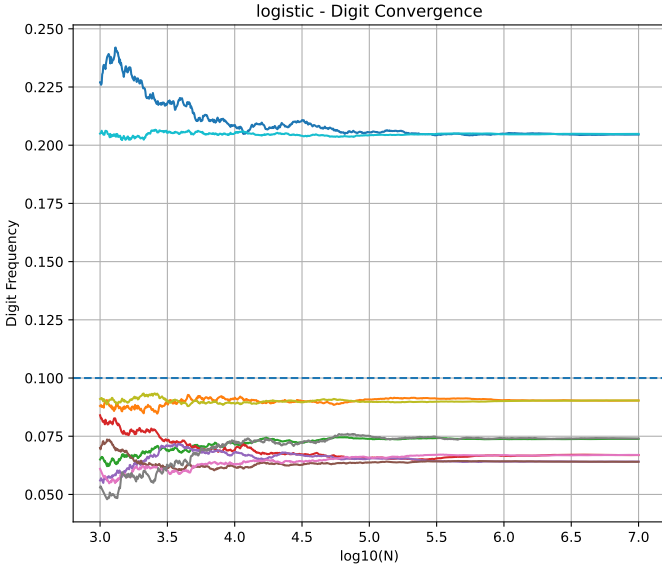
3.5.1 Digit Frequency Convergence



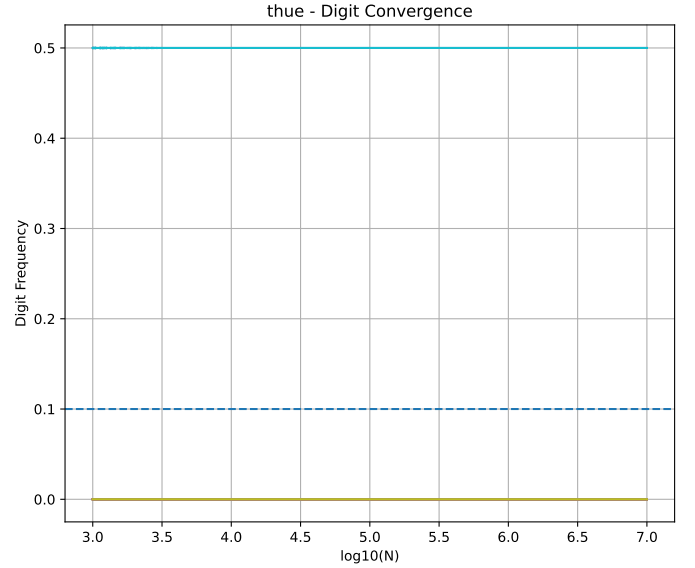
(a) De Bruijn [16]



(b) LCG [15]



(c) Logistic, $r = 4$ [17]

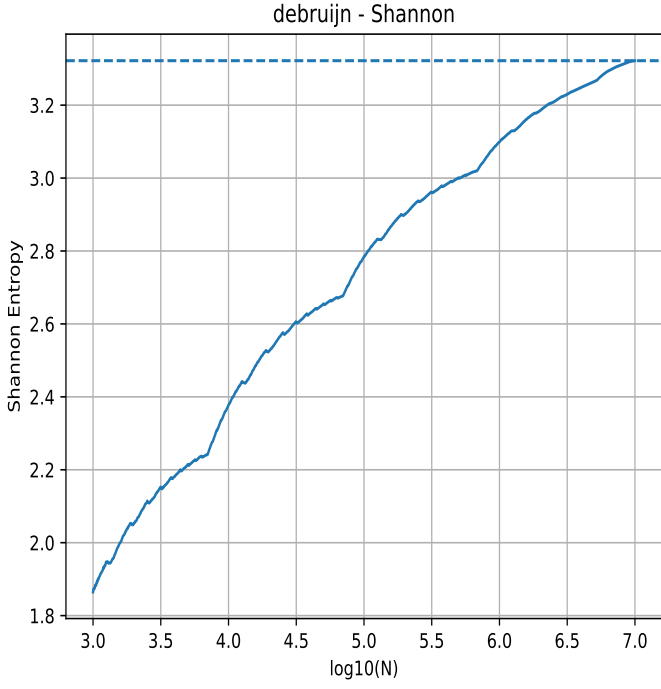


(d) Thue–Morse [18]

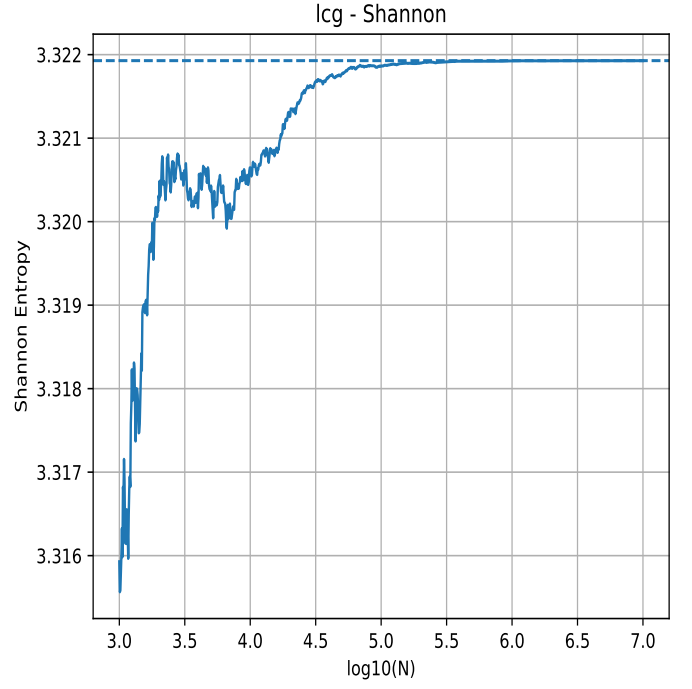
Figure 13: Digit frequency convergence for additional generators. Each curve represents one decimal digit (0–9). Colour mapping : blue = 0, orange = 1, green = 2, red = 3, purple = 4, brown = 5, pink = 6, gray = 7, olive = 8, cyan = 9. All sequences approach near-uniform frequencies at large N , though convergence patterns differ by construction.

All four generators exhibit approximate convergence toward uniform digit frequencies, though Thue–Morse and De Bruijn show structured oscillations reflecting their recursive and finite-cycle organization.

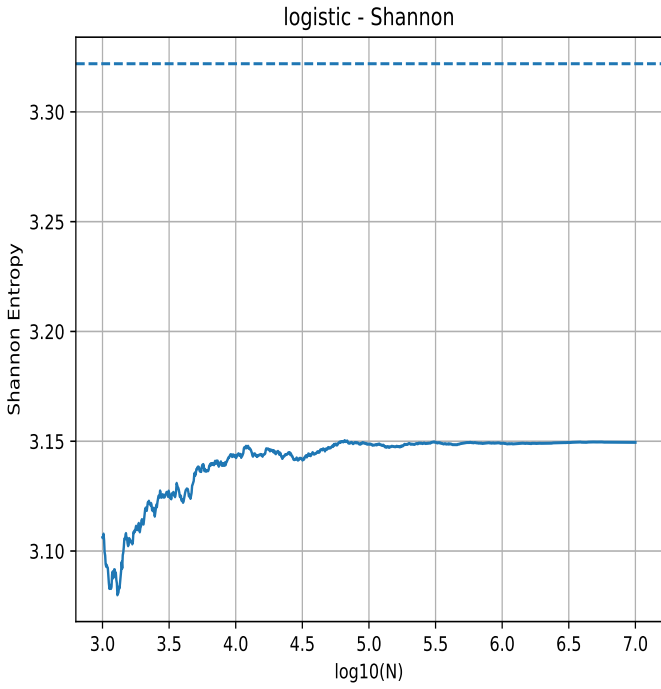
3.5.2 Shannon Entropy



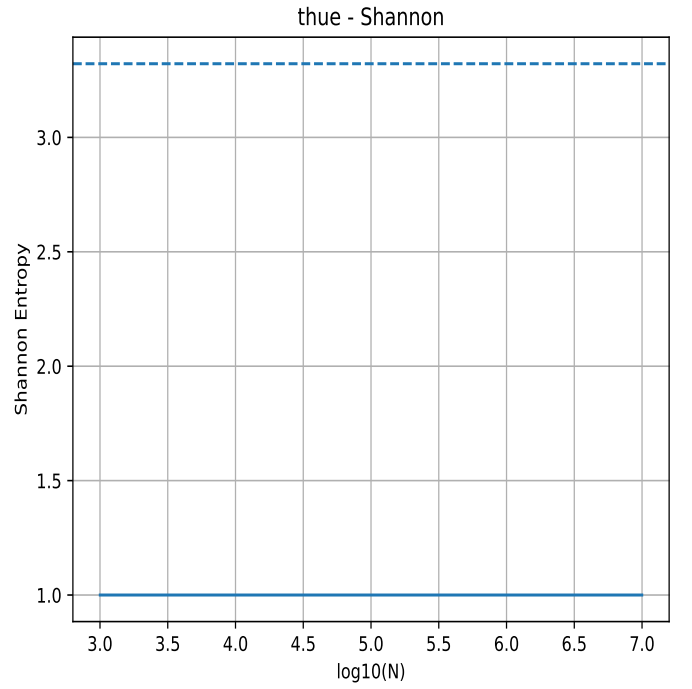
(a) De Bruijn



(b) LCG



(c) Logistic, $r = 4$

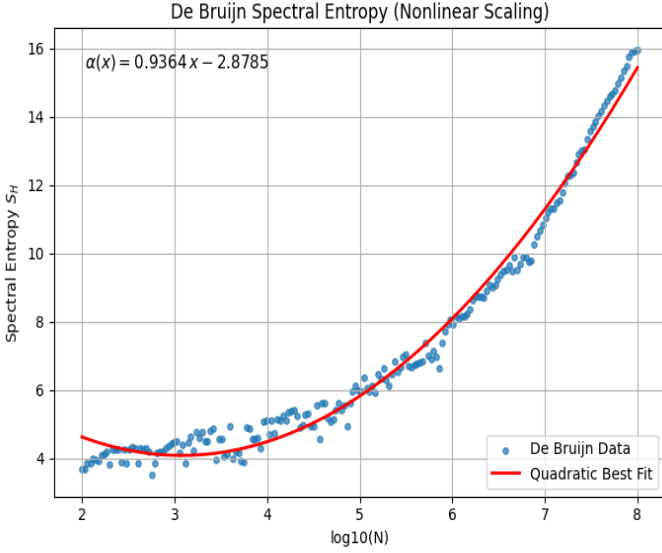


(d) Thue-Morse

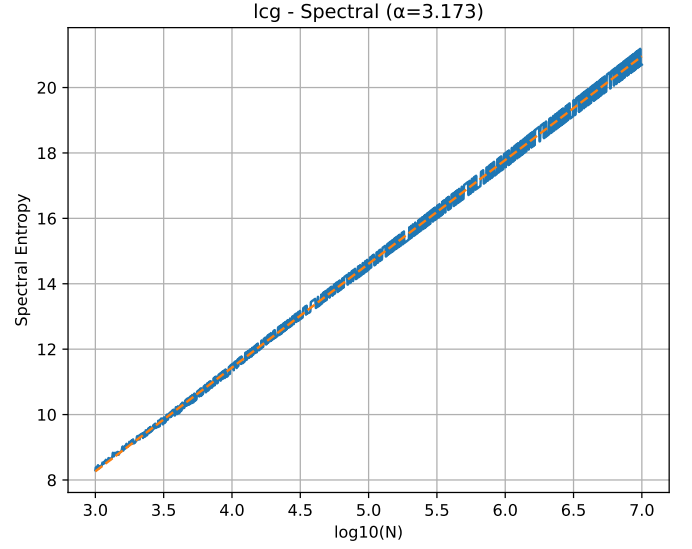
Figure 14: Shannon entropy for additional generators. All saturate near $H_{\max} = \log_2 10 \approx 3.322$ bits, confirming that Shannon entropy detects digit-level uniformity but not multiscale structure.

Consistent with Section 3.2, Shannon entropy rapidly saturates once digit frequencies stabilise, rendering deterministic, chaotic, and pseudorandom constructions indistinguishable [10].

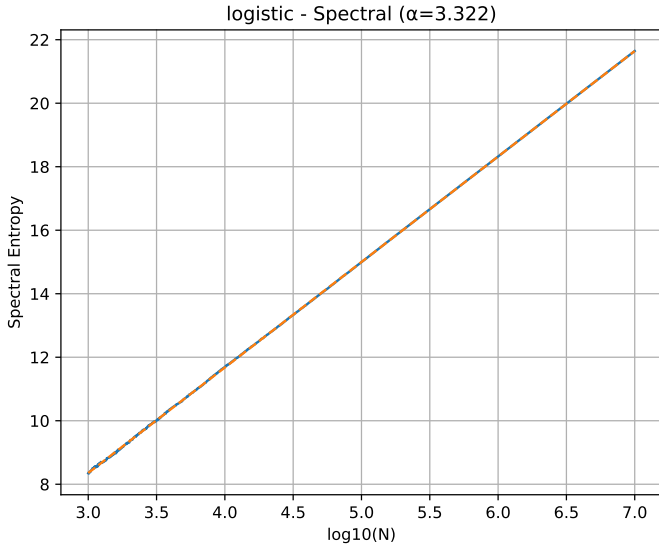
3.6 Spectral Entropy



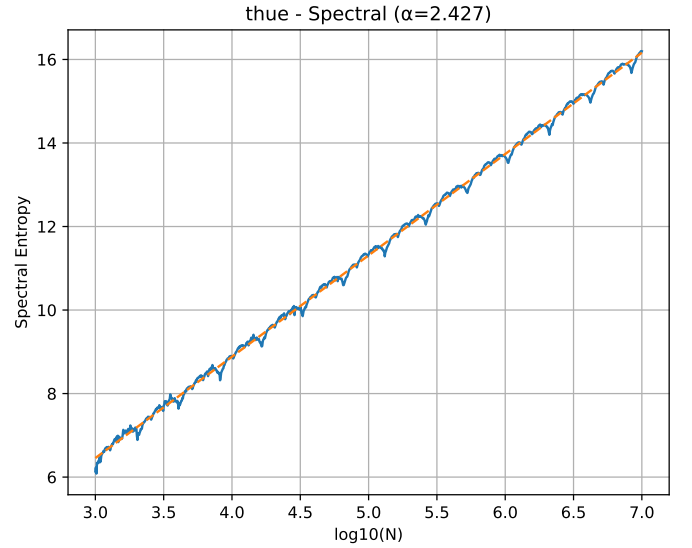
(a) De Bruijn ($\alpha_{\text{eff}}(N) \approx 0.936 \log_{10} N - 2.88$)



(b) LCG ($\alpha \approx 3.17$)



(c) Logistic, $r = 4$ ($\alpha \approx 3.32$)



(d) Thue–Morse ($\alpha \approx 2.43$)

Figure 15: Spectral entropy scaling for additional generators (continued). The De Bruijn linear fit is a poor description due to curvature (see Section 3.6.1).

In contrast to Shannon entropy, spectral entropy reveals substantial structural differences:

- **Logistic map** ($r = 4$ [17]): $\alpha \approx 3.32$, matching theory (Eq. 2), consistent with rapid decorrelation in the fully chaotic regime.
- **LCG** [15]: $\alpha \approx 3.17$, slightly reduced, consistent with known lattice correlations [19].
- **Thue–Morse** [18]: $\alpha \approx 2.43$, suppressed by long-range substitution correlations.
- **De Bruijn** [16]: pronounced convex curvature indicating nonlinear scaling; a quadratic fit captures the growth behaviour more accurately than any linear summary (Figure 15a). See Section 3.6.1.

3.6.1 De Bruijn Spectral Entropy: Scale-Dependent Complexity

The De Bruijn sequence is the only construction in our benchmark whose spectral-entropy growth is qualitatively nonlinear. We use a base-10 sequence of order $n = 7$, with cycle length $L = 10^7$ containing every 7-digit sub-string exactly once [16]. All data satisfy $N \leq 10^6 < L$, so we always sample a sub-cycle portion.

Raw spectral entropy. Figure 16(a) shows visible upward curvature in S_H vs. $\log_{10} N$, unlike the clean linear scaling of π , e , $\sqrt{2}$, and Random (Figure 9).

Quadratic fit. A quadratic model

$$S_H(x) = ax^2 + bx + c, \quad x = \log_{10} N, \quad (6)$$

captures the acceleration. Least-squares regression yields:

$$S_H(x) = 0.4682x^2 - 2.8785x + 4.4243 \quad (7)$$

with $R^2 = 0.991$ over $10^2 \leq N \leq 10^6$ (Figure 16b). Fit coefficients: $a = 0.4682 \pm 0.008$, $b = -2.8785 \pm 0.06$, $c = 4.4243 \pm 0.1$. The positive leading coefficient $a > 0$ confirms accelerating entropy growth.

Scale-dependent effective exponent. Differentiating Eq. (7):

$$\alpha_{\text{eff}}(N) = \frac{dS_H}{d\log_{10} N} = 2ax + b = 0.9364 \log_{10} N - 2.8785. \quad (8)$$

Evaluating at representative scales (Figure 16c):

- $N = 10^2$: $\alpha_{\text{eff}} \approx -1.01$
- $N = 10^4$: $\alpha_{\text{eff}} \approx 0.87$
- $N = 10^6$: $\alpha_{\text{eff}} \approx 2.74$
- $N = 10^8$ (extrapolated): $\alpha_{\text{eff}} \approx 4.61$

The effective exponent increases monotonically with scale. Within the tested range ($N \leq 10^6$), α_{eff} remains below the white-noise benchmark $\alpha_{\text{theory}} = 3.322$. Extrapolating Eq. (8), α_{eff} would reach α_{theory} near $N^* \approx 10^{6.6}$, though this lies outside the fitted range and should be treated with caution.

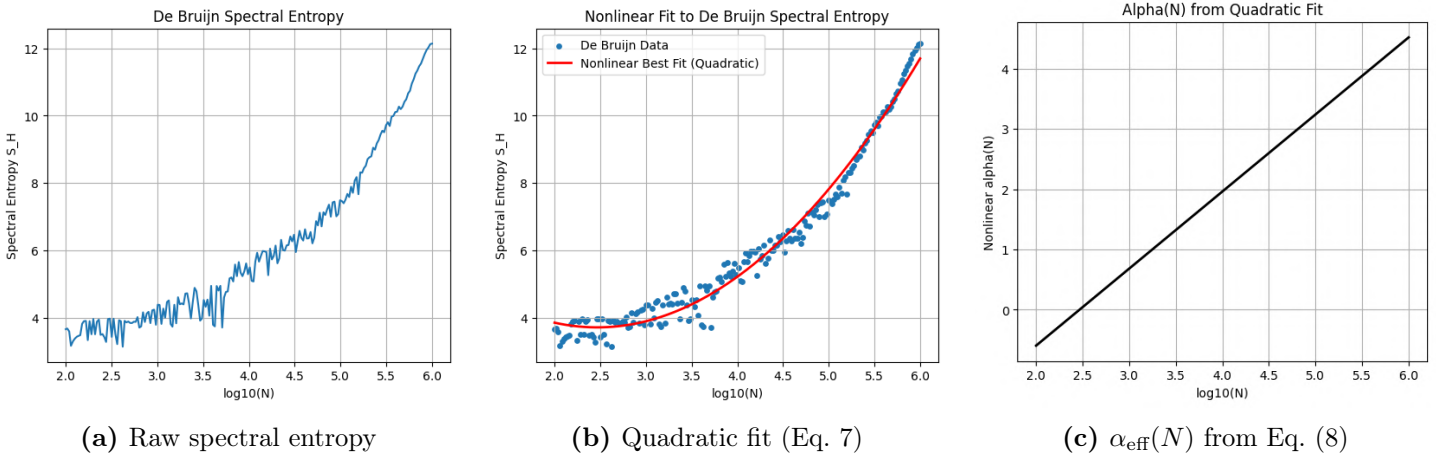


Figure 16: De Bruijn spectral entropy analysis. (a) Raw S_H vs. $\log_{10} N$ showing convex curvature. (b) Quadratic fit $S_H = 0.4682x^2 - 2.8785x + 4.4243$, $R^2 = 0.991$. (c) Effective exponent $\alpha_{\text{eff}}(N) = 0.9364 \log_{10} N - 2.8785$, increasing monotonically.

Physical interpretation. The acceleration reflects the De Bruijn sequence’s finite-cycle structure [16]. At $N \ll L = 10^7$, only a fraction of the cycle is sampled and strong periodic correlations concentrate energy in few modes. As N approaches L , combinatorial completeness forces all 7-grams to appear, progressively populating new Fourier modes faster than the logarithmic rate expected for an uncorrelated sequence (Eq. 5).

That the extrapolated α_{eff} exceeds α_{theory} beyond the tested range does not violate the absolute bound $S_H \leq \log_2(N/2)$; it would reflect a transient in the *local slope*—the sequence “catching up” from a suppressed-entropy regime, activating modes faster than the asymptotic rate. We expect that

for $N \gg L$, periodic repetition would collapse α_{eff} back toward zero, though this regime is not testable at $N_{\text{max}} = 10^6 < L$.

This contrasts with all other sequences tested: π , e , $\sqrt{2}$, and Random show constant $\alpha \approx 3.32$ (Table 1); Champernowne and Thue–Morse show constant but suppressed α ; only De Bruijn exhibits a scale-dependent α .

Summary of extended benchmark. The full hierarchy established by Tables 1 and the extended results is:

$$\underbrace{\text{Random / Chaotic}}_{\alpha \approx 3.32} \rightarrow \underbrace{\text{LCG}}_{\alpha \approx 3.17} \rightarrow \underbrace{\text{Thue–Morse}}_{\alpha \approx 2.43} \rightarrow \underbrace{\text{De Bruijn}}_{\alpha_{\text{eff}} \text{ variable}} \rightarrow \underbrace{\text{Champernowne}}_{\alpha \approx 1.24} \rightarrow \underbrace{\text{Periodic}}_{\alpha \approx 0}.$$

Spectral entropy thus distinguishes not only random from deterministic systems but also *among* deterministic structures, and can identify sequences whose complexity is itself scale-dependent.

4 Discussion

4.1 Quantitative Scaling Parameters

Table 1 collects the fitted spectral-entropy exponents for all ten sequences tested in this study.

Table 1: Spectral entropy scaling $S_H = \alpha \log_{10} N + \beta$ over $10^3 \leq N \leq 10^6$. The theoretical benchmark $\alpha_{\text{theory}} = \ln 10 / \ln 2 \approx 3.322$ (Eq. 2) assumes uniform Fourier mode population. Confidence intervals (95 % CI) are from weighted least-squares. Sequences are ordered by decreasing α .

Sequence	Normality status	α	95 % CI	R^2	Interpretation
Theory (Eq. 2)	–	3.322	–	–	Uniform modes
Logistic ($r = 4$)	N/A ^e	3.32	± 0.02	0.999	Matches theory
Uniform Random	Almost surely ^a	3.32	± 0.02	0.998	Matches theory
e	Conjectured ^b	3.32	± 0.02	0.998	Matches theory
π	Conjectured ^b	3.32	± 0.02	0.997	Matches theory
$\sqrt{2}$	Open ^c	3.32	± 0.01	0.999	Matches theory
LCG	N/A ^e	3.17	± 0.02	0.997	Slight lattice deficit
Thue–Morse	N/A ^e	2.43	± 0.03	0.994	Substitution correlations
De Bruijn	N/A ^e	α_{eff} variable ^f	–	0.991	Scale-dependent
Champernowne	Proven ^d	1.24	± 0.05	0.982	Block-structure deficit
Periodic	Not normal	0.00	–	–	Trivial structure

^aBy Borel’s theorem [1], almost all reals are normal; a uniformly drawn random sequence is normal with probability 1.

^bConjectured but unproven [2]. ^cOpen problem [7, 8]. ^dProven by Champernowne [6]. ^eNormality not applicable (generator output, not constant). ^fQuadratic fit: $\alpha_{\text{eff}}(N) = 0.9364 \log_{10} N - 2.8785$ (Eq. 8); ranges from ~ -1.01 at $N = 10^2$ to ~ 2.74 at $N = 10^6$ (within the tested range). R^2 is for the quadratic model of Eq. (7).

Key observations: The results establish a graded spectrum of spectral-entropy scaling across ten representative constructions. Sequences lacking detectable long-range correlations exhibit $\alpha \approx 3.32$, in quantitative agreement with the theoretical benchmark. Sequences with recursive, lattice, or block structure exhibit progressively reduced scaling.

- Five sequences— π , e , $\sqrt{2}$, Uniform Random, and the logistic map at $r = 4$ [17]—all yield $\alpha \approx 3.32$ with $R^2 > 0.997$, matching the theoretical prediction of Eq. (2).
- The LCG ($\alpha \approx 3.17$) shows a small but measurable deficit consistent with its known lattice structure [19], while Thue–Morse ($\alpha \approx 2.43$) exhibits a larger reduction reflecting long-range substitution correlations [18].
- Champernowne, despite proven normality [6], shows $\alpha \approx 1.24$, and the periodic control yields $\alpha = 0$.

- Together, these ten sequences form a graded hierarchy from trivial ($\alpha \approx 0$) through structured ($\alpha \approx 1\text{--}2.5$) to white-noise-like ($\alpha \approx 3.32$).

4.2 Normality vs. Spectral Complexity

The central observation from Table 1 is that, within the tested range ($10^3 \leq N \leq 10^6$), digit normality status does not predict spectral-entropy scaling. These two properties appear empirically independent at the examined scales.

Table 2: Normality–spectral-complexity matrix. Champernowne [6] occupies the “normal but spectrally structured” cell; generators occupy a separate row since normality is not defined for their outputs. All α values from Table 1.

	High scaling ($\alpha \gtrsim 3$)	Moderate scaling ($2 < \alpha < 3$)	Low scaling ($\alpha < 2$)
Proven normal	–	–	Champernowne
Conjectured normal	π, e	–	–
Almost surely normal	Random	–	–
Unknown normality	$\sqrt{2}$	–	–
Non-normal	–	–	Periodic
Generator (N/A)[†]	Logistic, LCG	Thue–Morse	De Bruijn [‡]

Normality conjectured but unproven for π and e [2]. [†]Normality is defined for real-number constants; generator outputs are classified by spectral scaling only. [‡]De Bruijn has scale-dependent α_{eff} : ~ -0.07 at $N = 10^3$ and ~ 2.74 at $N = 10^6$ (Eq. 8).

Champernowne demonstrates that proven normality does not guarantee high spectral complexity at the tested scales. High spectral scaling spans multiple normality categories and multiple origins, reinforcing the apparent independence of the two properties within the tested range.

4.3 Why Does Champernowne Show Reduced Scaling?

The concatenation $1|2|3|\dots|10|11|\dots$ [6] introduces systematic correlations: digits are organized by integer magnitude classes, digit “1” is over-represented in $[10^k, 2 \times 10^k)$ (peaks $\sim 40\%$; Figure 4), and position determines the current integer. These concentrate spectral energy in low-frequency modes at characteristic scales $T_k \sim 10^k$:

$$R_{uu}(\tau) \sim \sum_k A_k \cos(2\pi\tau/T_k), \quad (9)$$

consistent with the observed $\alpha \approx 1.24$ (Table 1). The entire sequence is generated by an $O(\log n)$ program [14].

4.4 Potential Applications

Spectral entropy could complement standard PRNG test suites (DIE-HARD [4], TestU01 [5]), which focus on local statistics: a generator passing chi-square tests but showing $\alpha \ll 3.32$ would indicate exploitable spectral correlations, as illustrated by the LCG’s $\alpha \approx 3.17$. Similarly, the Champernowne result ($\alpha \approx 1.24$ despite proven normality) suggests spectral tests could complement frequency-based assessments for quasi-random sequences used in Monte Carlo methods.

4.5 Limitations and Future Work

Our 1.5×10^6 digits yield high R^2 (Table 1) but represent a modest sample by modern standards [2]. Extension to $10^8\text{--}10^9$ digits would test scaling persistence and narrow confidence intervals. At small N the DFT has few independent modes and the spectral-entropy estimator may be biased; fitting only over $10^3 \leq N \leq 10^6$ mitigates this, but a systematic study of finite-size corrections would strengthen the results.

Comparing spectral entropy with alternative multiscale diagnostics—block entropy [10], permutation entropy, Lempel–Ziv complexity—would contextualise its discriminative power and is planned for future work. Planned base-2 and base-16 analyses will test whether spectral scaling is base-invariant, and expanding to additional constants (γ , ϕ , $\zeta(3)$) would probe the generality of $\alpha \approx 3.32$.

5 Conclusion

We have examined spectral entropy as a multiscale diagnostic for correlation structure in decimal digit sequences drawn from mathematical constants and algorithmic generators. While Shannon entropy rapidly saturates for all sequences with uniform digit frequencies, spectral entropy reveals systematic differences in scaling behaviour.

Sequences including π , e , $\sqrt{2}$, uniform random digits, and the fully chaotic logistic map exhibit scaling consistent with the theoretical prediction $\alpha_{\text{theory}} = \ln 10 / \ln 2 \approx 3.322$, indicating uniform Fourier mode population within the tested range. Structured constructions such as Champernowne, Thue–Morse, and periodic sequences show progressively reduced scaling, while the De Bruijn sequence exhibits scale-dependent growth.

These findings indicate that spectral entropy resolves correlation structure beyond digit-frequency statistics and provides a continuous hierarchy of complexity across deterministic, recursive, pseudorandom, and chaotic systems. Although the analysis is limited to $N \leq 1.5 \times 10^6$, the observed scaling behavior suggests that spectral entropy may serve as a complementary diagnostic in pseudorandom-number testing, symbolic complexity analysis, and multiscale structure detection.

Further investigation at larger scales and across additional bases and constants will be required to determine the universality and asymptotic robustness of the observed scaling regimes.

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Data Availability

All code and numerical data used to generate the results and figures in this study are publicly available at:

<https://github.com/ayushshukla652-sketch/spectral-entropy-digit-analysis>